

Assignment

Submission Instructions:

1. Write your Full Name, Student-ID, Theory Section on top of the first page
2. Your answer should be **handwritten**
3. Submit online here: <https://forms.gle/Vw5WzChNUaSV2FiNA>
4. **NO LATE SUBMISSION WILL BE ACCEPTED.**
5. Check the **section deadline mentioned** in discord.

Q1. [3+3 Marks] There are two types of data independence in database management systems: Logical Data Independence, and Physical Data Independence. Given the scenarios below, Identify what type of data independence both are and Explain your answer briefly (at max 3 sentences) with reference to Three-Schema Architecture.

- You work for a social media company. The company has decided to move from cloud hosting to physical hardware to support the server due to increase in demand.
- You work for a social media company. The company has decided to track the number of hours a user has stayed on their platform. This tracked_hours is not visible to the user but kept in the database.

Q2. [7 Marks] You are tasked with designing a Video Streaming System. The system must meet the following requirements:

- Each user will have a unique code, full name, email, and mobile number.
- Users can watch videos, which have a unique video ID, duration, upload_date, and play count.
- A video can be watched by many users and a user can watch many videos.
- Users can also comment on videos. Whenever a comment is made, the timestamp is recorded. A single user can comment multiple times.
- All videos must have comments, but not all users will leave a comment on the videos.
- Users can also follow other users.
- Some users might create a playlist. Each playlist has a playlist_name, tag, and a list of videos. Playlist_name is not unique, rather it is kept track of with the user's username and playlist_name combined.
- Users add videos to their playlist. One user can add many videos to their single playlist.

Now, draw an **ER diagram** of this system. Show at least 1 composite-multivalued attribute. Write down all assumptions (if any).

Q3. [7 Marks] You are tasked with designing a Driver Management System. The system must meet the following requirements:

- Drivers will have a unique DriverID, full name, email, and license number.
- Each vehicle has a unique VehicleID, model, capacity, and manufacturer.
- Each driver can operate many vehicles, but each vehicle can only be operated by one driver at a time.
- There are only 2 types of drivers: Full-time and Part-time. A driver can be both at the same time. Full-time drivers have a contract_type, while part-time drivers have working_hours.

- Some Vehicles are categorized by their type, such as Sedan, SUV or Electric. A vehicle can belong to only one type. Electric vehicles have BatteryCapacity, SUVs have cargoCapacity.
- A Part-time Driver can rent one Sedan at a time. When a Sedan is rented, the timeLimit is stored.
- A Full-time Driver can request repair from Staff on an SUV or an Electric. The issueDate is recorded when a repair request is made.
- A Staff will have StaffID, FullName, JoiningDate. The experience of a Staff can be calculated from JoiningDate.

Now, draw an **EER** (Extended Entity Relationship) diagram of this system.
Write down all assumptions (if any).
